PRESENTATION

2025年ICM-E题真题分析

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题目介绍

introduction

01

题目介绍

ICM-E题总介绍

01

E题聚焦环境、生态、可持续发展，强调跨学科建模与现实政策落地 。主题多围绕生态系统管理、资源可持续、气候变化影响、环境政策评估等

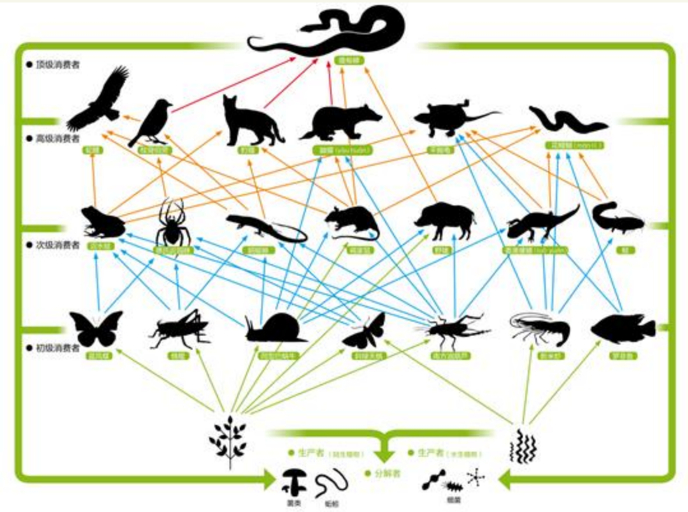
02

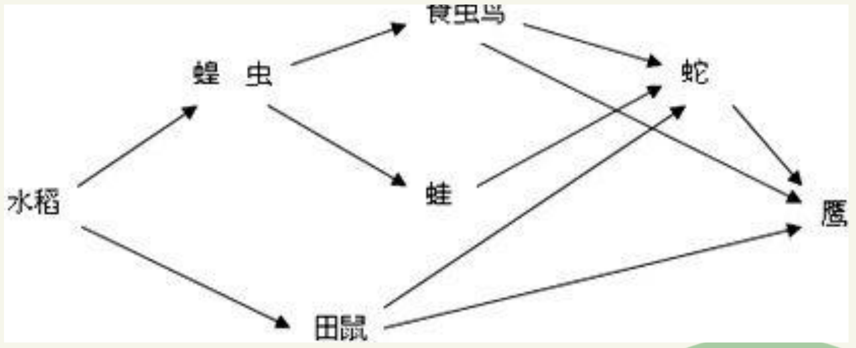
核心要求：用数学模型结合生态学、经济学、政策科学等，解决复杂环境与社会交叉问题，强调系统思维、多目标优化、可行性与社会价值。
能力考察：跨学科整合、动态系统建模（如系统动力学、微分方程）、数据驱动分析、方案评估与政策建议。

2025ICM-E题介绍

01 背景

- 生态系统的转变
- 土壤与农业问题
- 人类干预与生态失衡
- 新型食物网的形成
- 农业生态系统的行成过程





2025ICM-E题介绍

02 题目要求

1.建立一个动态演变模型，模拟从森林被清理后，到形成全新农田生态系统的整个过程。

2.模型必须包含两大驱动因素：

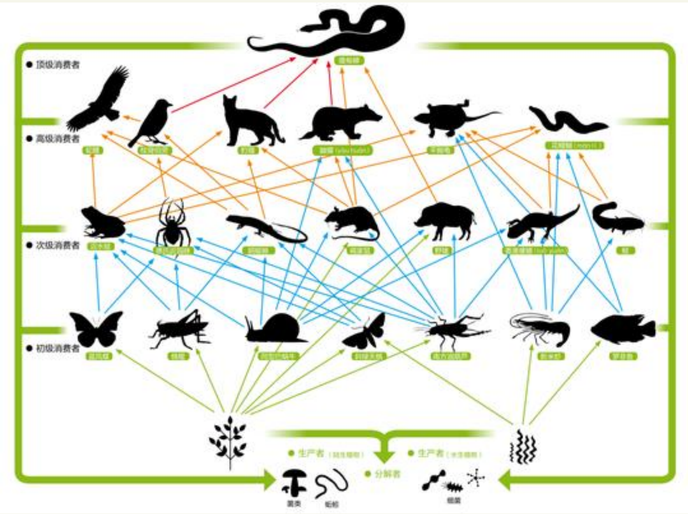
- 自然生态过程（如物种迁入、演替、食物网重构、季节性变化）。
- 人类决策因素（如作物选择、耕作方式、化学品使用策略）。

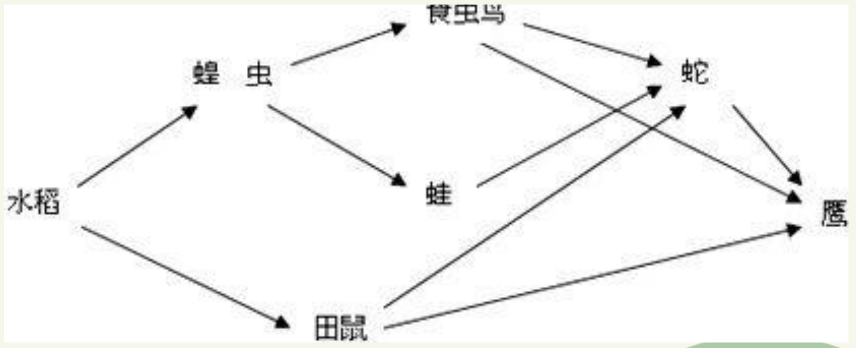
3.透过模型分析三个关键层面：

- 物种重新引入（如蝙蝠、鸟类）对农业生态平衡的影响。
- 化学投入（除草剂、杀虫剂）的增减策略，对产量、生态与经济效益的综合影响。
- 转向有机或部分有机种植，对系统可持续性、稳定性及可行性的潜在影响。

4.最终产出两项实用成果：

- 面向农民的政策与具体管理建议。
- 一封简短的信件，为农民转向更绿色种植方式提供科学依据与鼓励。





真题分析

analysis

02

建模要点

抓住要点

自然过程

人类决策

自然过程

一、建模要点

1．搭建农业生态系统基础食物网，明确生产者、消费者的层级与相互作用

2．纳入农业循环季节性，分析其对系统动态的长期影响

3．量化除草剂、杀虫剂对植物、昆虫、蝙蝠／鸟类种群及生态稳定性的影响

4．模拟本土物种回归过程，分析2个物种与现有环境的相互作用及对生态系统的改变

自然过程

二、数据来源

1．农业生态相关研究论文（食物网结构、物种间作用数据）
2．政府农业部门统计数据（农业循环周期、化学品使用量）
3．生态监测机构报告（本土物种种群变化、栖息地成熟速率）
4．环境科学数据库（化学品对生物种群的影响系数）

人类决策

一、建模要点

1．停用除草剂场景：从生产者、消费者视角分析生态系统稳定性变化

2．构建蝙蝠参与的食物网模型，分析其作为食虫动物、传粉者的作用，对比另一物种的生态修复效果

3．设计不同有机耕作构成要素的场景，分析其对生态系统各组成部分的影响

4．探讨有机耕作在害虫防治、作物健康、生物多样性等维度的表现

人类决策

二、数据来源

1．农业经济数据库（除草剂使用成本、有机耕作投入产出数据）
2．生态保护研究（蝙蝠及其他益生物种的生态功能数据）
3．有机农业案例报告（不同耕作模式的生态与经济效益）
4．可持续农业统计数据（生物多样性、长期产能变化）

构建步骤

1. 节点与关系定义（基础）

核心节点：生产者（作物、杂草）、初级消费者（植食昆虫）、次级消费者（捕食性昆虫、蝙蝠/鸟类）、分解者（土壤微生物）、干扰项（除草剂/杀虫剂）。

关系规则：能量流向（捕食/竞争/共生）；蝙蝠双效（捕食昆虫+传粉增益）；化学品按物种敏感度设差异化抑制系数。

工具：用NetworkX画有向图，标注作用类型与强度。

2. 核心方程建模（gLV扩展）

生产者（作物P）：$dP/dt = rP \cdot P(1-P/KP) - \alpha \cdot P \cdot H - \beta \cdot P \cdot I + \gamma \cdot B$（rP增长率，KP承载，H除草剂，I植食昆虫，B蝙蝠传粉增益γ）。

植食昆虫（I）：$dI/dt = rI \cdot I(1-I/KI) - \delta \cdot I \cdot C - \varepsilon \cdot I \cdot H + \phi \cdot P$（C捕食昆虫，ε化学品抑制，ϕ取食系数）。

蝙蝠（B）：$dB/dt = rB \cdot B(1-B/KB) - \sigma \cdot B \cdot H + \eta \cdot I$（η捕食收益，σ化学品影响）。

加入季节性：r、K等参数用正弦函数$r(t)=r0 \cdot (1+A \cdot \sin(2\pi t/T))$模拟周期波动。

3. 参数标定（可信度关键）

基础参数：r（增长率）、K（承载力）查生态数据库/文献；交互系数（α/β/δ）用实测数据拟合或专家打分+熵权法修正。

化学品系数：参考EPA、FAO毒性报告，设梯度（如除草剂对昆虫抑制ε=0.3，对作物α=0.1）。

蝙蝠增益：传粉γ取0.05–0.1，捕食η取0.2–0.4（依作物类型调整）。

构建步骤

4. 空间扩散扩展（可选进阶）
用Fick定律补空间项：$\partial P/\partial t = D \cdot \nabla^2 P$ + 原gLV项（D扩散系数），模拟物种迁移与化学品扩散。
工具：Python+SciPy求解偏微分方程，GIS可视化空间分布。
5. 模拟与验证
数值求解：用Runge-Kutta（RK4）法，步长dt=0.1，时间跨度5–10年。
验证指标：稳定性（计算平衡点与特征值，判断渐近稳定）；灵敏度（扰动关键参数α/γ，看种群波动幅度）；数据拟合（与实测种群数据$R^2 \geqslant 0.8$）。
6. 场景化适配（对接人类决策）
场景1（除草剂停用）：设H=0，对比生态稳定性与作物产量变化。
场景2（物种回归）：分别引入蝙蝠/益鸟，对比双物种对害虫抑制、传粉增益的差异。
场景3（有机转型）：调整施肥/耕作参数，用TOPSIS评估生态-经济综合效益。

论文分析

essay

03

论文总评

O奖

模型层层递进，跨学科融合好，数据扎实、可信度高，灵敏度分析聚焦关键参数。但部分假设简化过度，有机农业方案场景覆盖不全，整体基础扎实。

F奖

锚定双区域案例，模型拓展至分解者与土壤层，NSGA-III优化+TOPSIS多维度对比亮眼，实用性拉满。可惜模型复杂度高、普适性弱，物种回归覆盖典型物种偏少。

M奖

融入扩散方程补空间维度空白，遗传算法给出实操参数，高斯噪声测试抗干扰性，稳定性分析严谨。不足是AHP权重有主观性，物种与场景覆盖有限，普适性待提升。

H奖

构建DROPOFF框架+MLLV核心模型，逻辑闭环清晰，创新生态评级体系，量化物种差异化影响。但作物植物归为一类不符实际，微分方程过多易出错。

S奖

跨界引入CatBoost预测产量，熵权法三维度量化生态稳定性，12种物种构建完整食物网，场景丰富。但核心参数缺实测数据，未考虑分解者与呼吸作用，生态系统完整性不足。

论文总评

88%

共同优势

建模核心统一：均以Lotka-Volterra系列模型为基础，契合生态系统物种交互的核心逻辑，且均通过灵敏度分析验证模型稳健性；
任务覆盖全面：完整响应题目要求，涵盖食物网构建、物种回归影响、除草剂减量、有机农业转型四大核心任务；
落地导向明确：均通过给农民的信件转化理论模型为实践建议，兼顾生态效益与经济可行性。

81%

差异化亮点

模型深度：O奖、F奖在优化算法（NSGA-III、遗传算法）应用上更具优势；
维度拓展：M奖首次融入空间扩散因素，S奖创新引入机器学习预测，突破传统建模边界；
实践关联：F奖、M奖依托具体区域案例，结论更易落地；H奖、O奖则更侧重模型体系的严谨性。

89%

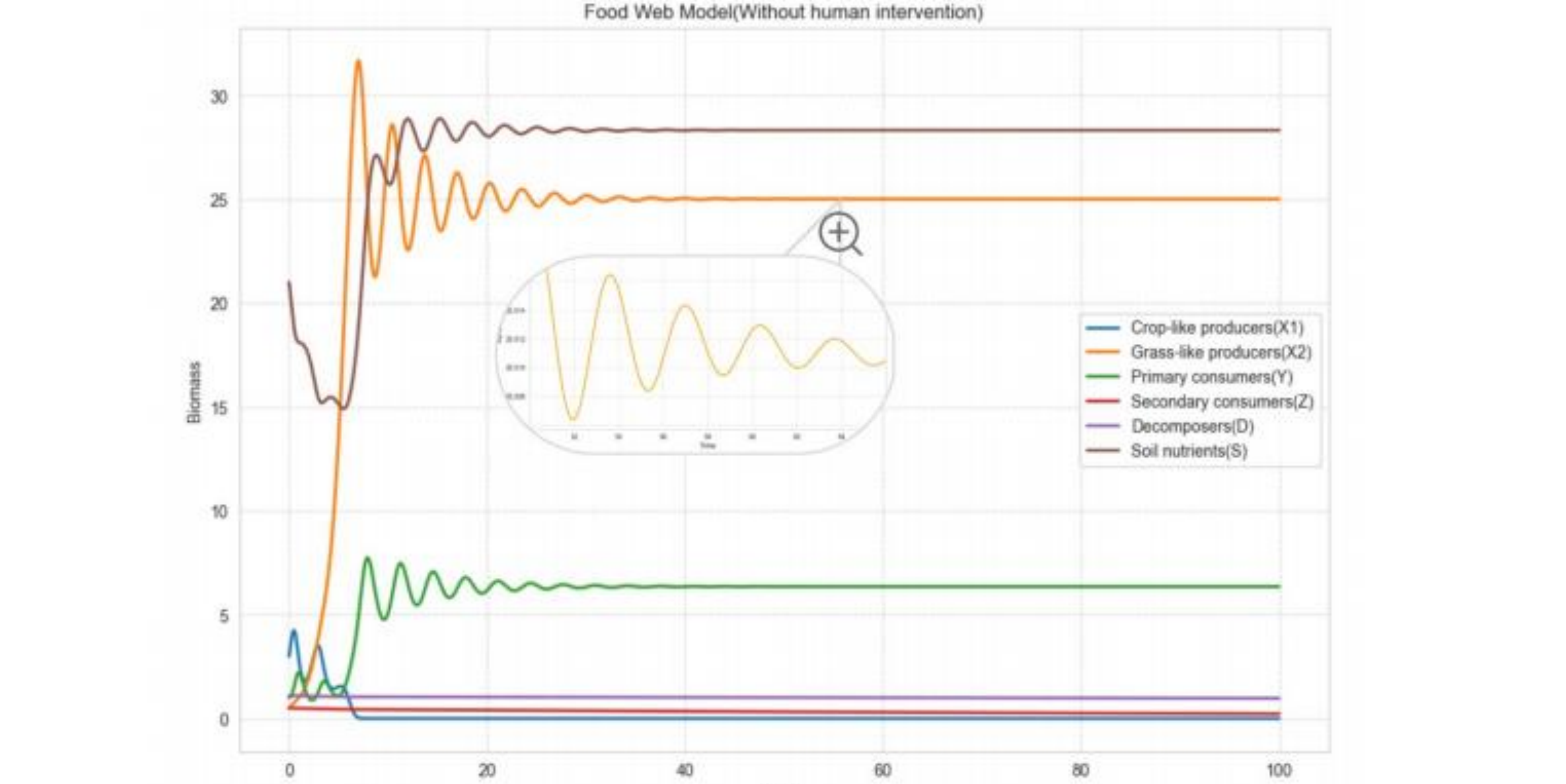
共同不足

假设简化普遍：均存在部分脱离实际的假设（如忽略物种内竞争、简化化学品作用机制）；
数据依赖度高：复杂模型需大量实测数据支撑，部分参数（如物种交互系数）缺乏权威参考标准；
场景覆盖局限：有机农业方案、物种回归类型的覆盖范围有限，未充分考虑不同气候、地理条件的差异

论文分评

O奖

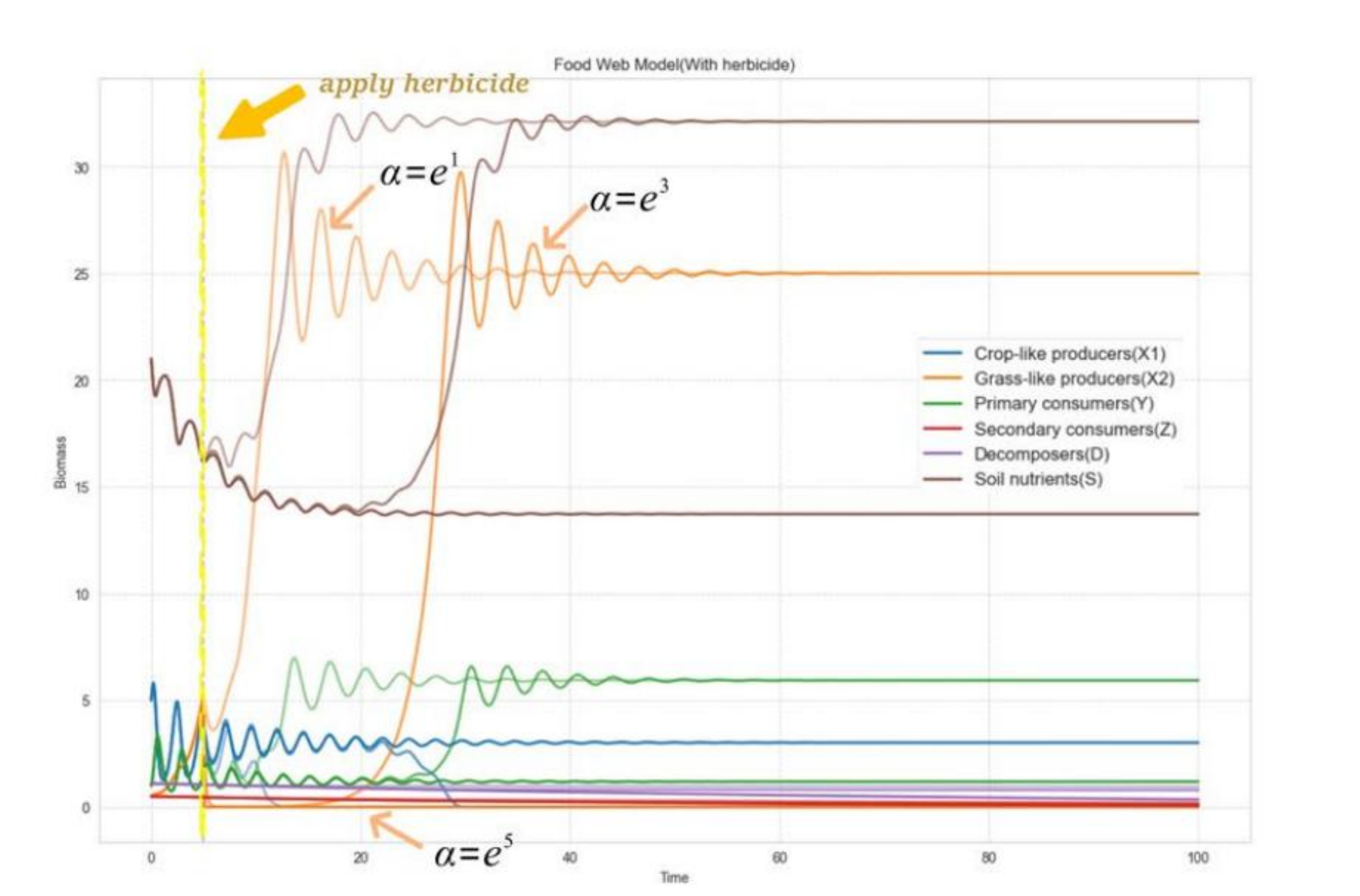
1．模型递进清晰：从三营养级食物链→食物网→含农业周期的食物网，层层深化；



论文分评

O奖

2．跨学科融合到位：整合微分方程、图论网络、EWM-AHP决策模型，覆盖生态、数学、管理多维度；3．数据支撑扎实：明确引用权威数据源，灵敏度分析聚焦关键参数（除草剂强度a)，结论可信度高



论文分评

O奖

1．部分模型假设简化过度（如假设杀虫剂对所有昆虫均有杀灭作用）；

2．有机农业方案仅聚焦3种措施，场景覆盖不够全面

In this context, we propose three solution pathways to achieve green agriculture. Then we combine them to obtain seven schemes, as well as a blank control group. Based on the set of differential equations in Problem 1, we change the initial assignments or coefficients of the equations, which can obtain the biomass of each species under the steady state of the ecosystem in different solutions. From this, a series of evaluation indexes such as Shannon index can be calculated. After the data were normalized, we applied the EWM-AHP to obtain the respective weights of the evaluation indicators, and then calculate the scores of different solutions. The weight setting in AHP is subjective; therefore, the EWM is employed to capture the variability within the data. The objectivity of the weights calculated through EWM effectively compensates for the limitations of AHP. Before applying the model, we carry out necessary preparation.

论文分评

F奖

1．区域案例具象：以加州中央谷地、亚马逊雨林为研究对象，增强模型实用性；

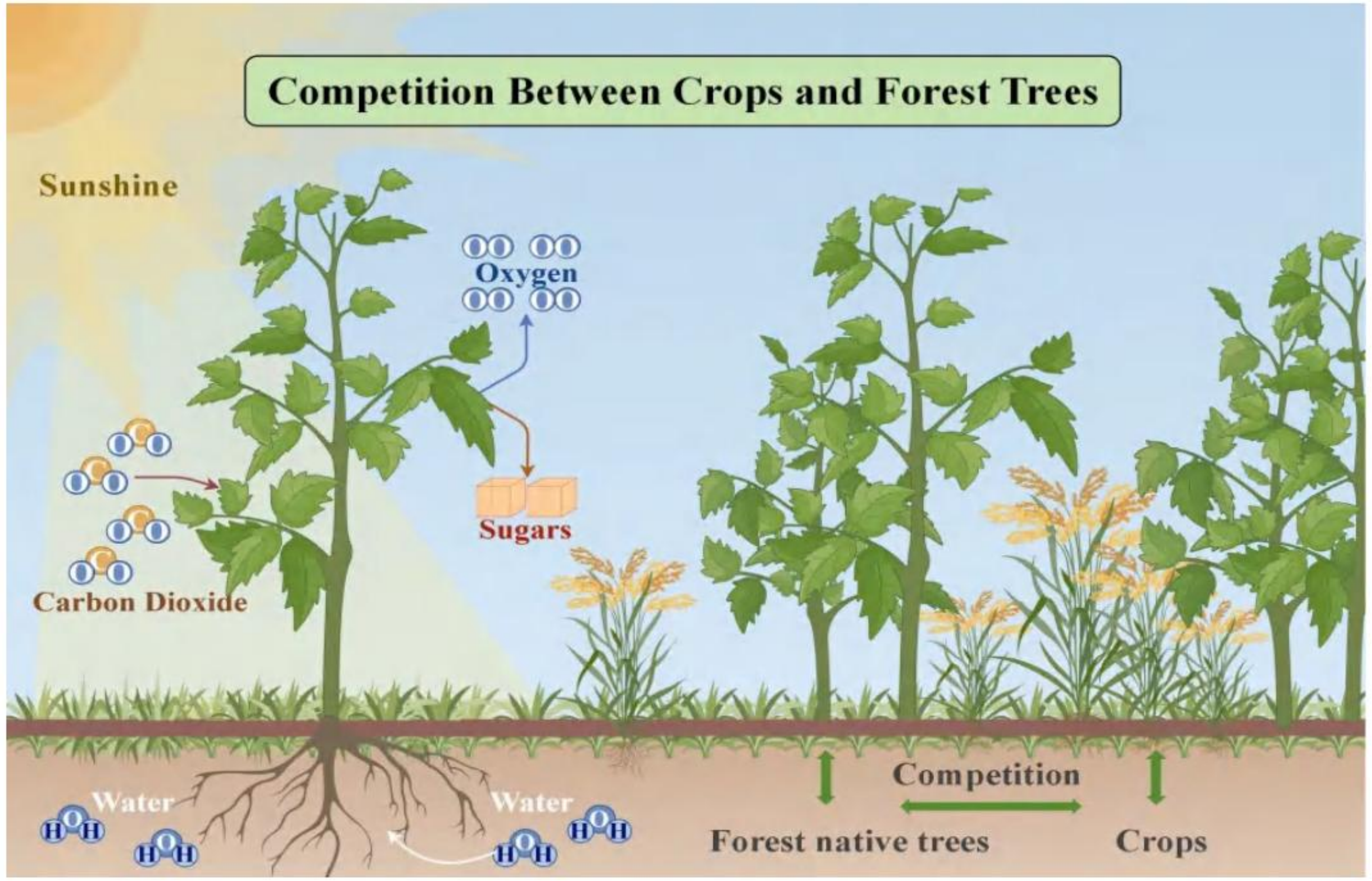


Figure 3: Crops and forest trees compete for sunlight, nutrients and water

论文分评

F奖

2．模型创新拓展：将LV/gLV模型延伸至分解者、土壤有机质，引入NSGA-III算法优化多目标规划；

3．对比维度丰富：用TOPSIS法系统对比传统农业与有机农业的生态、经济、社会效益

Table 2: The Result of Planning

Year	Pure Organic Strategy			Mixed Strategy		
	Soil Health	Biodiversity	Changes in Production	Soil Health	Biodiversity	Changes in Production
Year1	60	0.55	5200	60	0.55	5500
Year3	68	0.68	6100	65	0.59	6400
Year5	76	0.72	7200	71	0.62	7600
Year7	81	0.77	7800	75	0.65	8500
Year10	88	0.85	8100	77	0.68	9000

Summarizing the above results, we can introduce the following strategy for organic farms:

- If the goal of the farm is to ensure long-term ecological priority for thirty to fifty years, choose a purely organic strategy.
- If seeking economic output in the short term of ten to twenty years, choose a mixed strategy.
- Avoid traditional inorganic strategies that harm the land while also failing to ensure long-term economic output.

论文分评

F奖

1．模型复杂度高，依赖大量数据输入，普适性受限；

2．物种回归仅聚焦蝙蝠、老虎，未覆盖更多典型本土物种

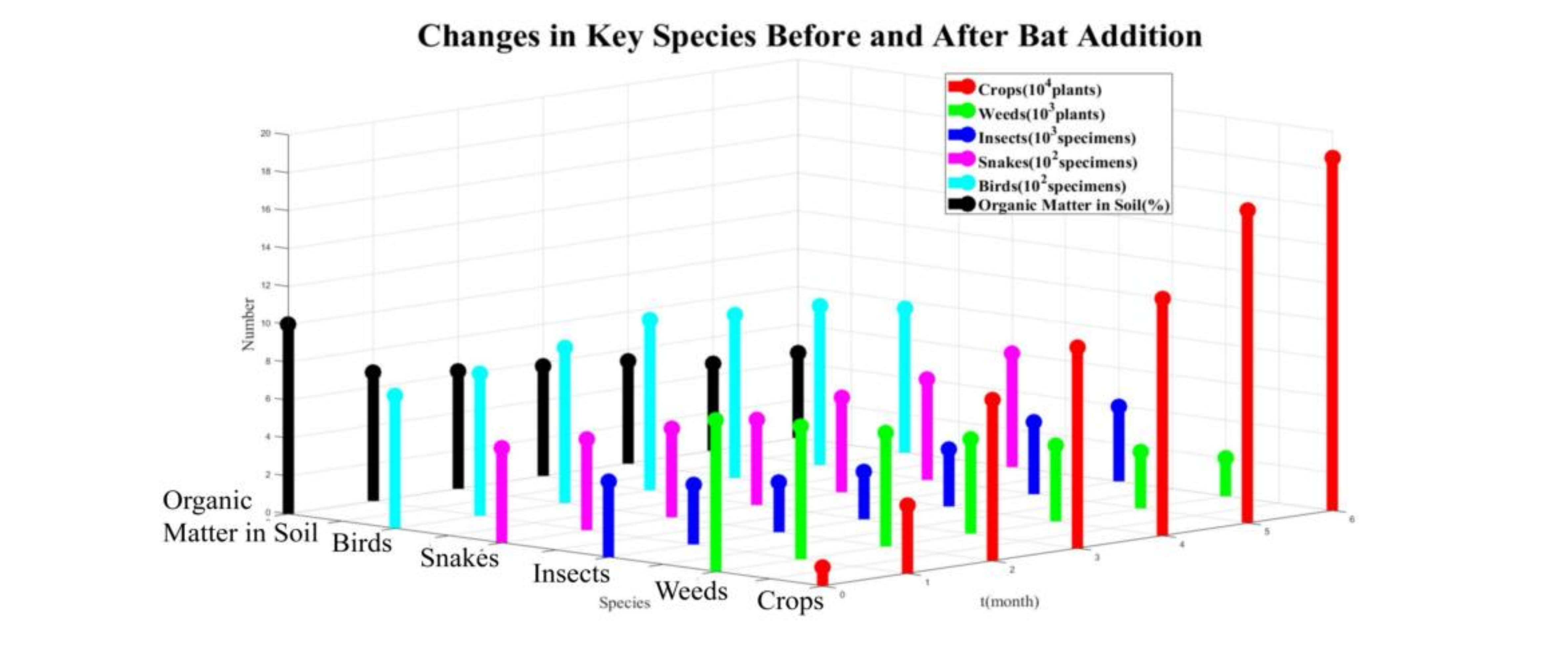


Figure 13: The Impact of the Bats Addition

论文分评

M奖

1．空间因素突破：融入扩散方程（Fick定律），模拟化学品扩散与物种空间迁移，填补空间维度空白；

$$\frac{\partial C(x,y,t)}{\partial t} = D\left(\frac{\partial^2 C(x,y,t)}{\partial x^2} + \frac{\partial^2 C(x,y,t)}{\partial y^2}\right) - \lambda C(x,y,t) + S(x,y,t) \tag{5.1}$$

where λ and D are the volatilization and spraying coefficients and $S(x,y,t)$ is the source of the chemicals. Additionally, we assume that the cropland satisfies the Dirichlet boundary condition $C(x,0) = C(x,L) = C(0,y) = C(L,y) = 0$, which means the surrounding will absorb the chemicals we utilize. Let us calculate the evolution after homogeneous intermittent fertilization ($S = S_0\delta(t)$). To avoid the divergence of the Fourier series, we assume the initial condition at $t = 0$ is

$$C_0(x,y) = \begin{cases} C_0, 95\% \text{ interior area} \\ 0, 5\% \text{ edge area} \end{cases} \tag{5.2}$$

By solving the eigenvalues and eigenfunctions of the Helmholtz equation $C_{xx} + C_{yy} + kC = 0$, we can easily get the eigenvalues and the eigenfunctions $k_{mn} = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2, C_{mn} = \sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)$. According to the method of separation of variables, we assume the solution in the form of $C(x,y,t) = X(x,y)T(t)$ and substituting it into the original equation. The final solution of the chemical diffusion equation is:

$$C(x,y,t) = \sum_{n,m=1}^{\infty} C_{mn} e^{-(k_{mn}D+\lambda)t} \sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right) \tag{5.3}$$

$$C_{mn} = \frac{4(\cos 0.05m\pi - \cos 0.95m\pi)(\cos 0.05n\pi - \cos 0.95n\pi)}{nm\pi^2} C_0 \tag{5.4}$$

论文分评

M奖

2．优化工具实用：采用遗传算法优化人类决策参数，给出具体可操作的参数组合；

3．稳定性分析严谨：引入随机高斯噪声测试系统抗干扰能力

In the traditional Lotka-Volterra Model, the local asymptotic stability at the attractive point can be determined by the negative definiteness of the Jacobian matrix within a small perturbation expansion, owing to the autonomy of the equations[3-4]. However, the presence of time-dependent factors in our model, such as $Kf'(t)$ and $\phi_0 \sin \omega t$, complicates the judgment of system stability when crops encounter stochastic disturbances during growth. To numerically test the stability of the crop system, we introduce random Gaussian noise into the RKF45 iterator. For clarity in our comparison, we focus on the crop, wild weed, and first consumer components of the ecosystem.

Figure 7.1 simulates a special case 100 times with different levels of disturbances. The parameters of this case are delicately chosen to maintain a long-term stable Lotka-Volterra model when the crop factor is removed. Under small disturbances (shown in a)), the stability of the Lotka-Volterra model is still maintained. However, as the disturbances become larger (shown in c) and d)), the time-dependent term starts to have a significant effect, and the entire system gradually transitions into chaos.

论文分评

M奖

1.AHP权重确定存在主观性；

2．物种与场景覆盖有限，模型普适性待验证

8.2 Weakness

- **Subjectivity in AHP weights**:
  Similar to the problems associated with AHP in other studies, there is an element of subjectivity in our use of the AHP model to determine the weights of the evaluation indicators.
- **Lack of partial data**:
  The lack of reference food web data in the network, probably due to the difficulty of ecosystem data surveys, has led to a lack of reference for some of the model parameters to be fitted to the actual data.
- **Limited range of species and scenarios**:
  our study has a relatively limited number of species of interest in food web modeling and limited analysis of specific scenarios for organic agriculture. This limited scope limits the generalizability of our findings.

论文分评

H奖

1. 框架体系化：提出DROPOFF整体框架+MLLV核心模型，逻辑闭环清晰；

5 Development——The MLLV Model to Depict Species

The farmland is a native natural environment, as illustrated in Figure 4. In the initial stage of model development, since it was just the beginning phase of forest-to-farm argiculture, we simplified the food web to some extent, as illustrated in Figure 5. We refer to the starting state of the model as the “initial state.” During the process, we first modeled the relationships among the most fundamental species. When this model reached a stable state, we termed it the “final state.” This section focuses on studying the transitional model of species changes between the initial state and the final state.

Figure 4: Farm-to-Forest Agriculture

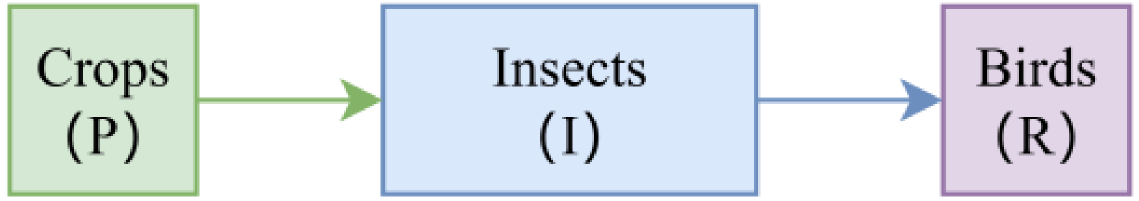


Figure 5: The Food Web Considered in the Initial Stage

论文分评

H奖

2．评估指标创新：构建生态系统评级体系（结合作物健康、稳定性、恢复力）；

3．对比分析深入：量化蝙蝠与传粉者对生态的差异化影响

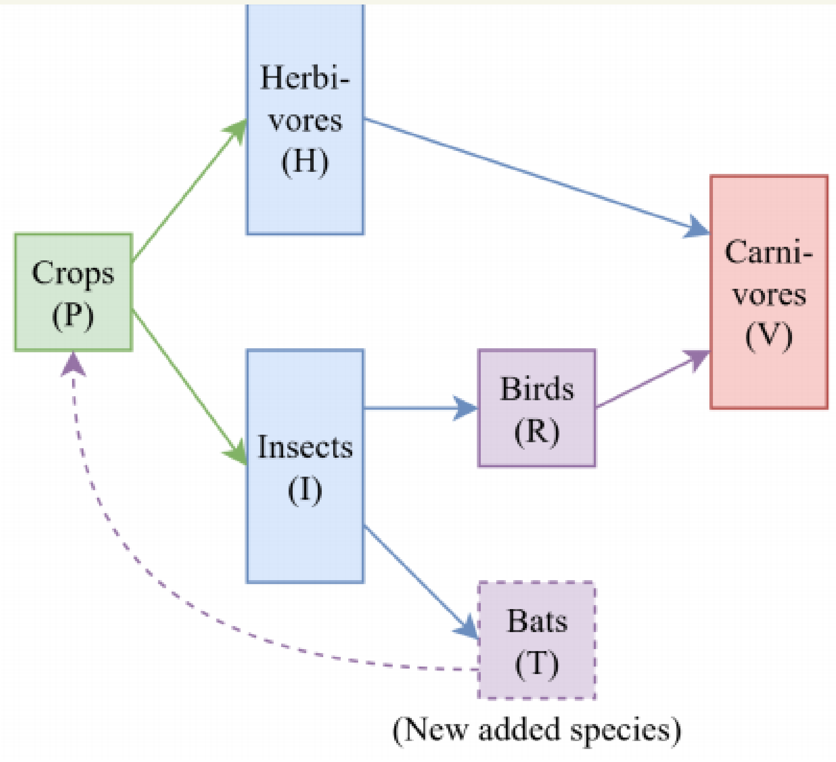


Figure 15: Main Food Web with Herbicides

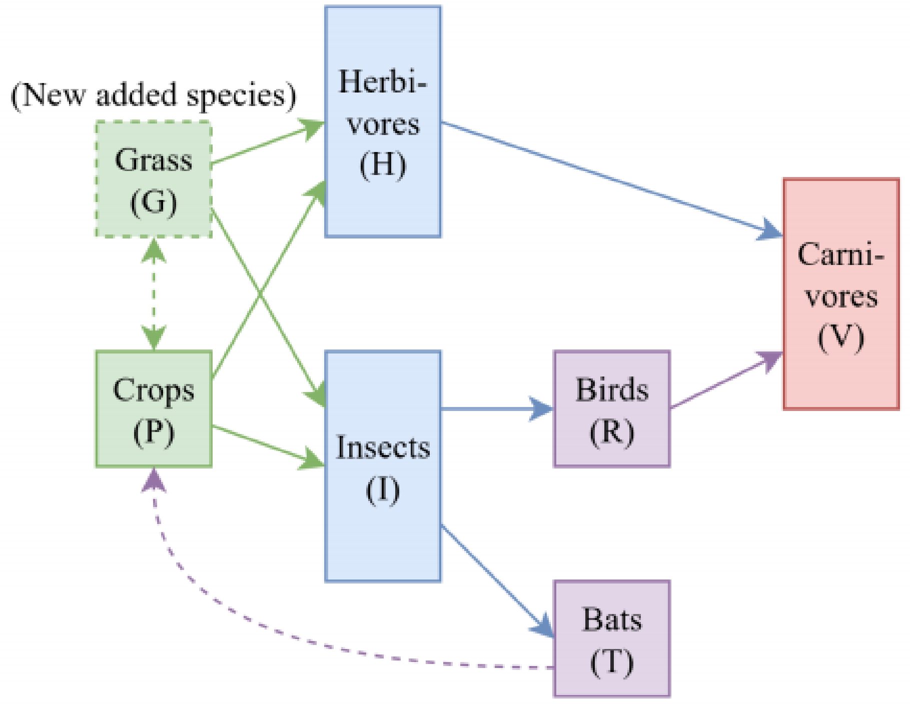


Figure 16: Main Food Web without Herbicides

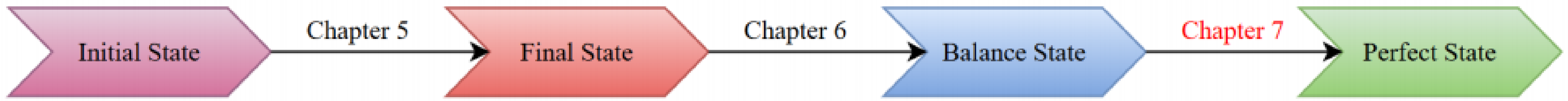


Figure 17: Schematic Diagram of the Stages Covered up to the Current Chapter

论文分评

H奖

1．假设作物与植物归为一类，与实际生态差异较大；

2．微分方程数量随研究深入增多，计算繁琐易出错

$$\begin{cases} \frac{\mathrm{d}dP(t)}{\mathrm{d}t} &= P(t)[r_P(1-\frac{P(t)}{K_P}) - |\alpha_{PI}|\cdot I(t)] \\ \frac{\mathrm{d}I(t)}{\mathrm{d}t} &= I(t)[r_I(1-\frac{I(t)}{K_I}) + |\alpha_{IP}|\cdot P(t) - |\alpha_{IR}|\cdot R(t)] \\ \frac{\mathrm{d}R(t)}{\mathrm{d}t} &= R(t)[r_R(1-\frac{R(t)}{K_R}) + |\alpha_{RI}|\cdot I(t)] \\ P(0) &= P_0 \\ I(0) &= I_0 \\ R(0) &= R_0 \end{cases} \tag{2}$$

$$\lambda^3 + A\lambda^2 + B\lambda + C = 0 \tag{11}$$

$$A = \mathrm{tr}(J) = J_{11} + J_{22} + J_{33} \tag{12}$$

$$B = J_{22}J_{33} - J_{23}J_{32} + J_{11}J_{33} + J_{11}J_{22} - J_{12}J_{21} \tag{13}$$

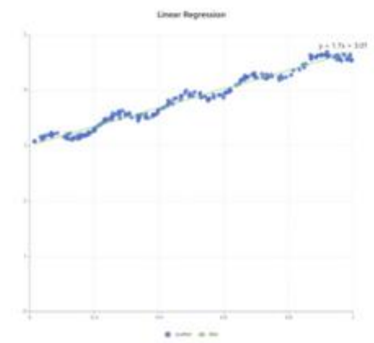
$$C = \det(J) = J_{11}J_{22}J_{33} - J_{11}J_{23}J_{32} - J_{12}J_{21}J_{33} \tag{14}$$

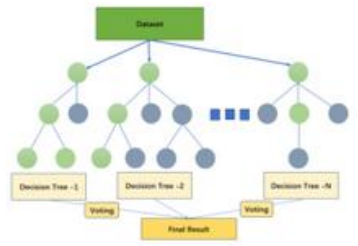
$$\begin{cases} \frac{\mathrm{d}P(t)}{\mathrm{d}t} &= P(t)[r_P(1-\frac{P(t)}{K_P}) - |\alpha_{PI}|\cdot I(t) - |\alpha_{PH}|\cdot H(t)] \\ \frac{\mathrm{d}I(t)}{\mathrm{d}t} &= I(t)[r_I(1-\frac{I(t)}{K_I}) + |\alpha_{IP}|\cdot P(t) - |\alpha_{IR}|\cdot R(t)] \\ \frac{\mathrm{d}R(t)}{\mathrm{d}t} &= R(t)[r_R(1-\frac{R(t)}{K_R}) + |\alpha_{RI}|\cdot I(t) - |\alpha_{RV}|\cdot V(t)] \\ \frac{\mathrm{d}H(t)}{\mathrm{d}t} &= H(t)[r_H(1-\frac{H(t)}{K_H}) + |\alpha_{HP}|\cdot P(t) - |\alpha_{HV}|\cdot V(t)] \\ \frac{\mathrm{d}V(t)}{\mathrm{d}t} &= V(t)[r_V(1-\frac{V(t)}{K_V}) + |\alpha_{VH}|\cdot H(t) + |\alpha_{VR}|\cdot R(t)] \end{cases} \tag{15}$$

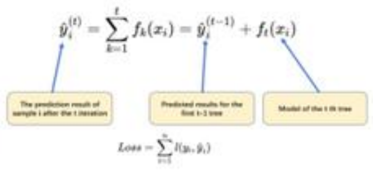
论文分评

S奖

1．技术跨界创新：首次引入机器学习(CatBoost模型）预测作物产量，提升决策科学性；







Algorithm 1: Ordered boosting

input : $\{(\mathbf{x}_k, y_k)\}_{k=1}^{n}, I$;

$\sigma \leftarrow$ random permutation of $[1, n]$;
$M_i \leftarrow 0$ for $i = 1..n$;
for $t \leftarrow 1$ **to** I **do**
  for $i \leftarrow 1$ **to** n **do**
    $r_i \leftarrow y_i - M_{\sigma(i)-1}(\mathbf{x}_i)$;
  for $i \leftarrow 1$ **to** n **do**
    $\Delta M \leftarrow$ $LearnModel((\mathbf{x}_j, r_j) : \sigma(j) \leq i)$;
    $M_i \leftarrow M_i + \Delta M$;
return M_n

Figure 8: Machine Learning models we used

Now the data is randomly divided into the test set and the training set, and the effects of the four models are compared. We use the square of the difference between the predicted yield and the actual yield in the test set as the basis for comparison.

And we find that the result of effect is as followed:

```
Linear Regression - MSE: 213.73
Random Forest - MSE: 131.51
XGBoost - MSE: 192.98
CatBoost - MSE: 124.94
```

Figure 9: model result

论文分评

S奖

2．稳定性评估量化：基于熵权法，从振幅、周期性、长期趋势三维度评估生态稳定性；

3．场景具象化：构建完整食物网（含12种典型物种），模拟多种物种引入、化学品使用场景

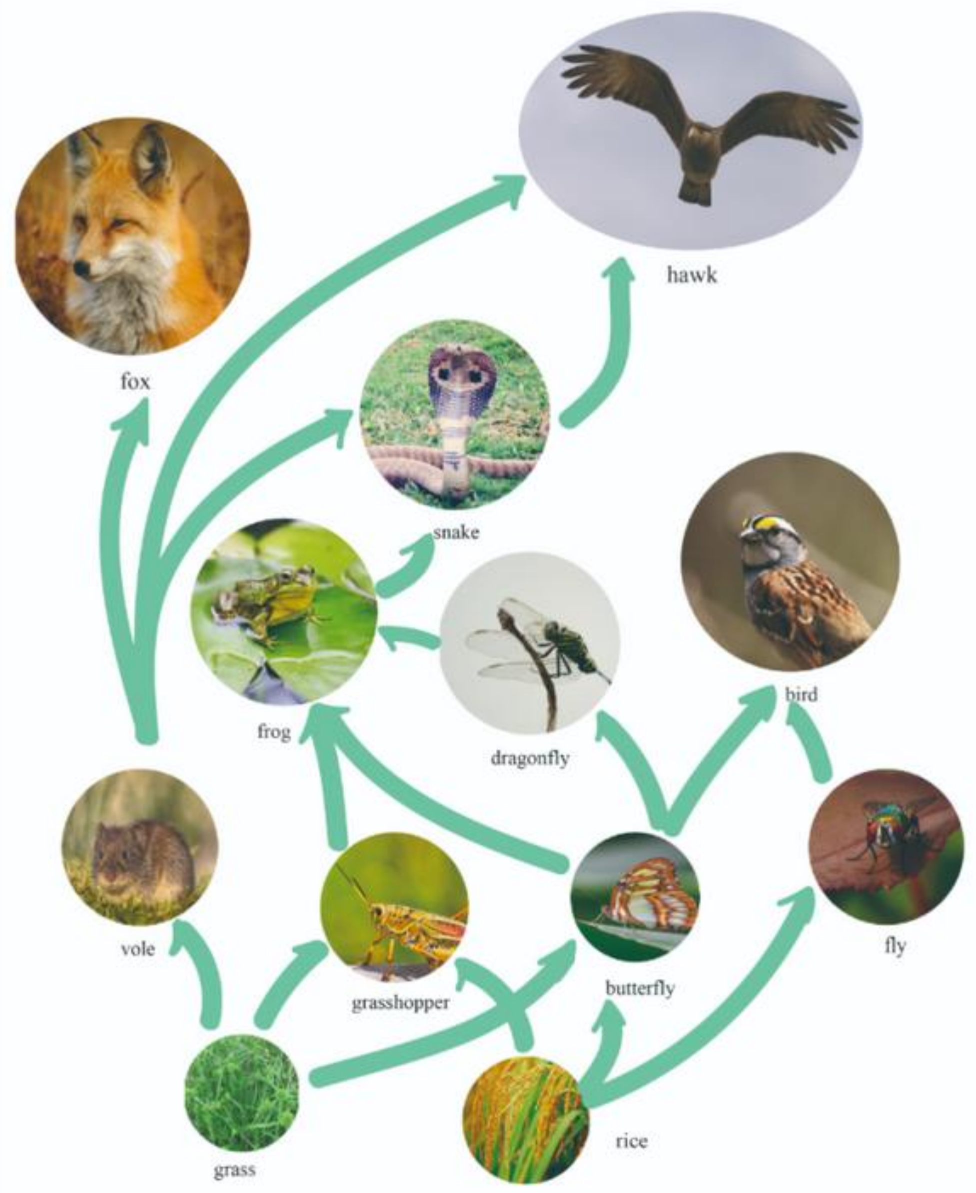


Figure 10: food web

论文分评

S奖

1．部分参数设置缺乏实测数据支撑（如物种季节出生率）；

2．未考虑分解者、呼吸作用，生态系统完整性不足

- **Our parameter settings lack data support.**. It's very difficult to find specific data online, such as the birth and death rates of voles in different seasons and the intensity of competition between rice and weeds. Therefore, it's challenging to validate our model with data. Although the final results generally conform to common sense, we could definitely do better in terms of details if there were abundant data.
- **Our ecosystem model doesn't take into account decomposers and respiration.**. The theory of these two factors are relatively complex. It's difficult to incorporate decomposers into the food web, and the theory of respiration is also quite complex and abstract. So, we didn't consider these two important factors, which may lead to significant differences in the simulation results.

应赛策略

strategy

04

应赛策略

1.题目理解与界定：开篇清晰拆解题目关键词，通过数据或背景分析界定问题优先级，抓住主要矛盾，抓住本质

Model and Analyze:

In places throughout the world, scenarios like this one occur. As a member of the Consideration of Mature Agricultural Practices (COMAP) group, you have been asked to construct a model to track habitat change from forest-to-farm. Your supervisor has given your team the lead in determining how a **converted forest area** can change over time as the ecosystem evolves along with accompanying agricultural choices. Your supervisor wants the analysis to include both ***natural processes*** as well as ***human decisions***. Therefore, you should start your model of a newly cleared converted forest area ecosystem and track the model through stages of change due to the changes in species in addition to the many impacts of farming practices. You can make assumptions to build a situation of forest-to-farm, or you can use data and information from stages in a real historic sample of this kind of evolution. You may want to consider the following in your analysis:

应赛策略

2.跨学科融合度：明确标注跨学科工具的适配性，问题E（环境类）融合“环境科学（污染扩散模型）+经济学（成本收益分析）+政策学（政策模拟评估）”，用污染扩散数据支撑政策制定的成本测算，不同学科方法形成互补（如用数学建模量化效果，用社会学调研验证模型合理性）。

Model and Analyze:

In places throughout the world, scenarios like this one occur. As a member of the Consideration of Mature Agricultural Practices (COMAP) group, you have been asked to construct a model to track habitat change from forest-to-farm. Your supervisor has given your team the lead in determining how a **converted forest area** can change over time as the ecosystem evolves along with accompanying agricultural choices. Your supervisor wants the analysis to include both ***natural processes*** as well as ***human decisions***. Therefore, you should start your model of a newly cleared converted forest area ecosystem and track the model through stages of change due to the changes in species in addition to the many impacts of farming practices. You can make assumptions to build a situation of forest-to-farm, or you can use data and information from stages in a real historic sample of this kind of evolution. You may want to consider the following in your analysis:

应赛策略

3.模型构建与合理性

1）假设明确且有依据（例：假设“人口增长率稳定”时，引用近5年人口数据支撑）；

2）模型选择说明理由（如用“系统动力学模型”而非“线性回归”，因需体现多因素动态反馈）；

3）推导过程清晰（含公式、变量定义，关键步骤有逻辑衔接）；

4）模型有鲁棒性分析（如改变参数取值，验证结果是否仍稳定）。

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应赛策略

4.数据支撑与分析：数据来自权威渠道（如世界银行、政府统计年鉴、学术论文数据库），标注来源和时间；数据处理有细节（如缺失值填补、异常值剔除的方法说明）；通过可视化（图表、热力图等）呈现数据趋势，且分析紧扣模型（例：用历史污染数据校准扩散模型参数，用模拟数据验证政策实施效果）。

Model and Analyze:

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应赛策略

5.结论与建议：结论直接回应题目核心诉求；建议分层次（如短期应急措施+长期规划），含具体实施路径；结合跨学科视角提出差异化方案（如环境问题同时给出技术解决方案和政策保障措施）。

Model and Analyze:

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应赛策略

6.结构与表达：遵循“问题重述→模型假设→模型构建→数据处理→结果分析→结论建议→优缺点与拓展”的经典结构；公式、图表有编号（如“图1 污染扩散趋势图”“公式（1）供需平衡方程”），注释清晰；语言简洁，无语法错误，专业术语使用准确（如区分“焓变”“熵变”的定义）。

Model and Analyze:

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THANK YOU FOR WATCHING

谢谢观看 汇报完毕

主讲人：郭怡嘉 龚诗雅 程阳